

ISSAI

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AI AVATAR

Virtual Learning Platform

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AI AVATARS

Initial goal

Python → working AI product, no prior coding experience required

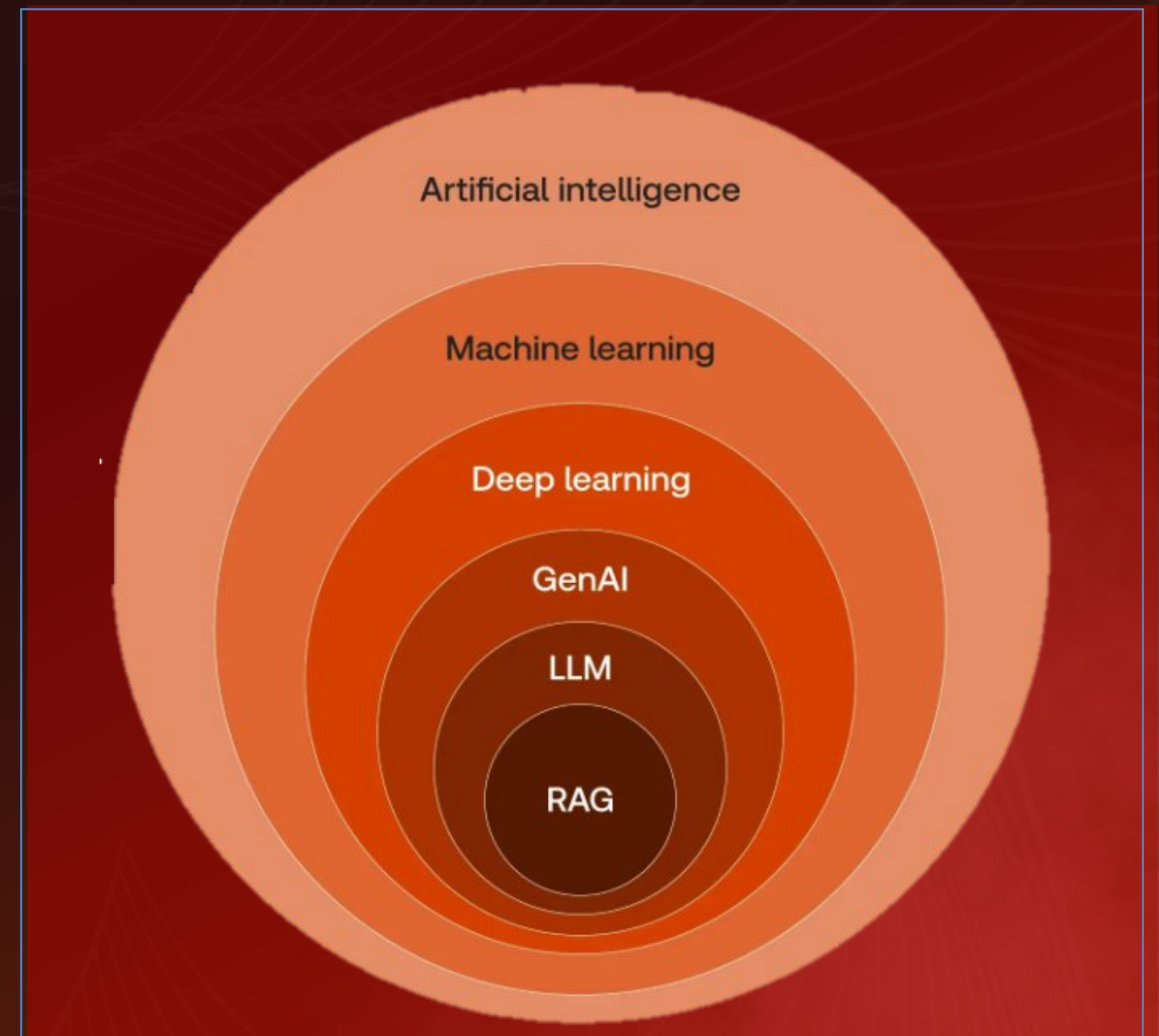
Goal: build a voice-interactive 3D AI avatar (Kazakh / Russian / English) that hears, thinks, and speaks

Required deliverables: working avatar, mini-research report, final live demo

Tech stack: Python, MangiSoz ASR/TTS, Oylan LLM, React, Three.js

Personal goal: go beyond a demo is to build a real, useful application

Outcome direction: an AI legal assistant for Kazakhstan, grounded in real law



Acquired knowledge and skills

Weeks 1 - 2

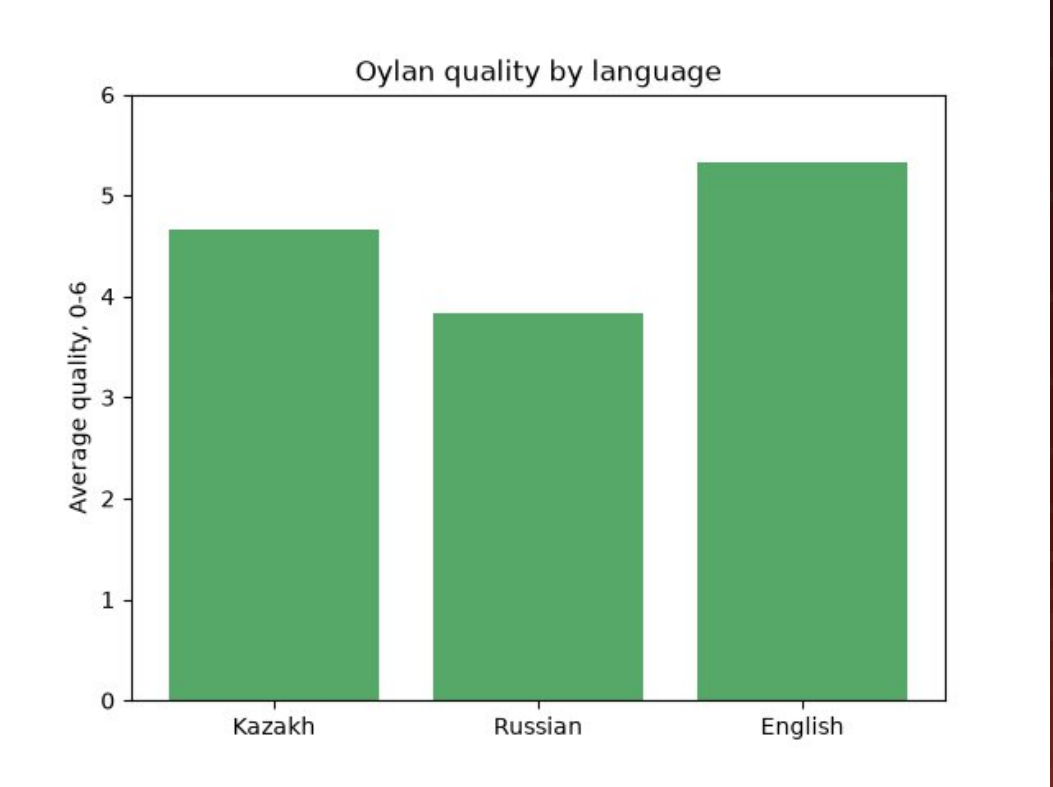
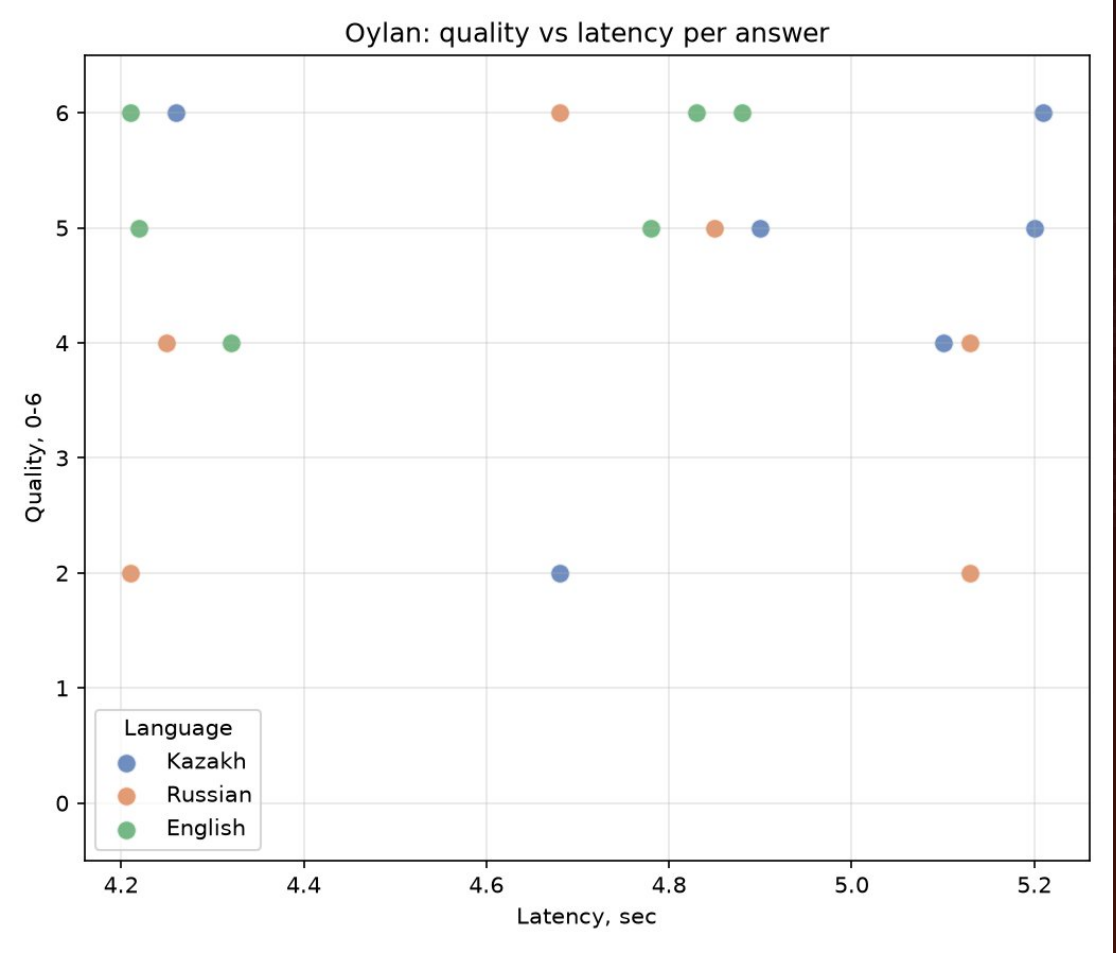
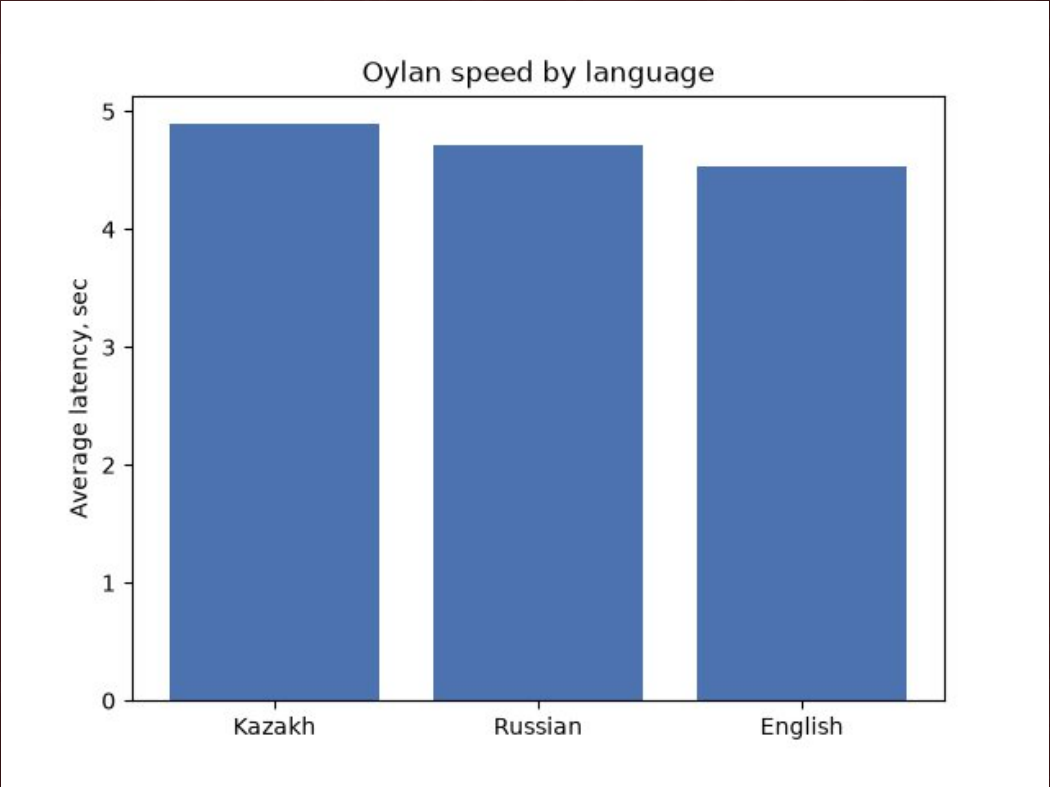
Python & Voice Pipeline

Python basics: variables, functions, loops, JSON
REST APIs, HTTP requests
Chained ASR → LLM → TTS into first voice prototype

Week 3

Prompt Engineering & Research

System prompts, few-shot examples, temperature
Designed AI "personalities" (tutor, guide, teacher)
Ran structured experiment: 10 questions × 3–5 prompt variants
Measured latency & quality, visualized with Matplotlib



Oylan Cross-Language Experiment: tested 20 physics questions across Kazakh, Russian, and English, each with and without a physics-teacher system prompt (120 answers total). Scored every answer on correctness, clarity, and language compliance.

Finding: English answers scored highest across all models (Oylan: 5.5/6 EN vs 4.2/6 RU vs 4.8/6 KK). Main quality bottleneck was response truncation, not factual errors — Oylan knew the right answer but often cut off mid-sentence, especially in Kazakh and Russian. System prompt did not improve quality (4.67 vs 5.00 without), likely because longer prompted answers hit the token limit more often.

Week 4

HTML/CSS/JS, React (components, state, hooks)
React basics: components, state, hooks.)
Three.js, GLB models, morph targets, visemes

Weeks 5 - 6

Web scraping & data engineering (adilet.zan.kz, egov.kz)
Vector embeddings & databases (ChromaDB, multilingual-e5)
Hybrid search: dense + BM25 + Reciprocal Rank Fusion
Cross-encoder reranking (BAAI/bge-reranker-v2-m3)
Formal evaluation methodology (Hit@1, Hit@5, MRR)
Backend service design (Flask microservice)

Acquired knowledge and skills

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AI AVATARS

ISSAI AvatarGavin

Avatar & scene controls

История консультации

Задайте вопрос о своих правах — ответ сошлётся на конкретную статью.

Response timing

No timing yet. Ask a question to see how the answer was retrieved and spoken.

Microphone icon

Нужно разрешения микрофона

Отправить →

GROUNDING

GROUNDING IN KAZAKHSTAN LAW · OYLAN LLM · VOICE: MALE · LANGUAGE: RU

ISSAI AvatarConnor

КР ЗАНДАРЫНА НЕГІЗДЕЛГЕН

Кеңес тарихы

Құқығыңыз туралы сұрақ қойыңыз — жауап нақты бапқа сілтенеді.

Response timing

No timing yet. Ask a question to see how the answer was retrieved and spoken.

Microphone icon

Микрофон рұқсаты қажет

Жіберу →

GROUNDING

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ISSAI AvatarKara

GROUNDING IN RK LAW

Consultation history

My landlord raised rent mid-contract without my agreement, is that allowed?

Generally, a landlord cannot raise rent mid-contract without the tenant's agreement. Most rental contracts specify the terms of rent increases, and any changes typically require mutual consent. If your contract does not allow for mid-term rent increases, your landlord may be in breach of contract. You should review your contract and consider consulting with a local housing authority or attorney to understand your rights.

Response timing

Speech → text0.00s

MangiSoz STT transcription

Legal search0.00s

RAG: retrieve articles from KZ law index

Legal answer0.68s

Groq / Oylan LLM response

Text → speech4.48s

MangiSoz TTS (parallel halves)

Total5.17s

End-to-end latency

PROCESSING BREAKDOWN

LLM13%

TTS87%

Mic permission needed

Send →

GROUNDING

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small_llm_avatar/README....

Auyeskhan_Alibekov_Track_...

Voices | ElevenLabs

ISSAI Avatar — Legal Comp...

Avaturn | Realistic avatar cre...

Google

ISSAI Avatar

Gavin

HA OCHOBE ЗАКОНОВ PK

История консультации

ИП на упрощёнке заработал 10 миллионов за полгода — какой налог и когда платить?



Response timing

No timing yet. Ask a question to see how the answer was retrieved and spoken.

Нужно разрешение микрофона

ИП на упрощёнке заработал 10 миллион

Ищу в законах...

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 **ISSAI Avatar**

Kara

● GROUNDED IN RK LAW



 **Consultation history**

Police took my phone during detention and won't return it, what law protects me?



Response timing

No timing yet. Ask a question to see how the answer was retrieved and spoken.

 Mic permission needed

Police took my phone during detention and won'

 Searching the law...

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 **ISSAI Avatar** | Connor

Avatar & scene controls

Consultation history



Ask about your rights — every answer cites a specific article.



Response timing

No timing yet. Ask a question to see how the answer was retrieved and spoken.

 Mic permission needed

when was the album

Send →

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**Thanks for
your attention**